K. SUCHIR REDDY

Software & Data Trainee Engineer · Systems & Analytical Engineering

📍 Bengaluru, Karnataka, India 📞 +91 7899597757 ✉️ ksuchirreddy@gmail.com 🌐 ksuchirreddy.github.io 💻 GitHub

🔗 LinkedIn

PROFESSIONAL SUMMARY

4th-year CS student at DSCE (CPI: 8.15) with strong fundamentals in data structures, algorithms, and modular Python/Java engineering. Practical background in ML benchmarking and relational databases.

EDUCATION

Dayananda Sagar College of Engineering (DSCE), Bengaluru — **CPI: 8.15 / 10.0**

Bachelor of Engineering (B.E.) — Computer Science & Business Systems — 4th Year Undergraduate (Batch 2026)

TECHNICAL SKILLS

Languages & Core CS: Python (Advanced), Java, C++, SQL, Data Structures & Algorithms

Analytics & ML: Pandas, NumPy, Scikit-learn, Power BI, PyTorch

Systems & Databases: MySQL, Flask, REST APIs, Git / GitHub, Linux

KEY ENGINEERING PROJECTS

Multi-Model Code Generation & Benchmarking Engine

`[Python, PyTorch, Transformers, AST (Abstract Syntax Trees), Evaluation Metrics]`

- Built an automated evaluation pipeline measuring LLM-generated code across runtime latency, memory, and syntactic correctness.
- Utilized Abstract Syntax Tree (AST) parsing in Python to validate grammatical code structures without execution overhead.
- Benchmarked multi-model inference performance across varied prompt complexities with structured JSON logging.

AI Content Moderator & Social Platform

`[Python, Flask, FastAPI, NLP, MySQL, Scikit-learn, REST APIs]`

- Engineered high-throughput REST APIs in Flask/FastAPI for automated real-time text toxicity classification and content filtering.
- Architected relational MySQL database schemas managing user profiles, flagged content queues, and audit logs.
- Built responsive web interfaces with complete frontend-to-backend data flow and validation.

Adaptive Traffic Signal Optimization using Computer Vision

`[Python, OpenCV, YOLO, Real-Time Video Streams, Dynamic Scheduling]`

- Developed real-time vehicle density detection from multi-camera feeds using OpenCV and YOLO to dynamically adjust green signal timing.
- Designed low-latency image processing pipelines reducing simulated intersection bottlenecks and waiting delays.
- Structured robust video stream preprocessing and bounding-box telemetry for dynamic signal phase allocation.

LEADERSHIP & ACADEMIC MENTORSHIP

AI Enthusiast & Peer Mentor (DSCE): Hosted hands-on mentoring sessions for engineering peers, guiding students through practical machine learning workflows, algorithms, and clean production code practices.